Research Article

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Extracting Inductive Bias Using Generative Models

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Abstract: The connectivity structure of a neural network influences how well it generalizes from limited data, yet designing this structure often requires domain expertise. We propose a generative multi-task learning framework that learns a distribution over network structures from solutions to related tasks. Our central hypothesis is that structural regularities shared across these solutions provide an inductive bias that transfers to new tasks. We generate structures autoregressively, specifying both connectivity and weight sharing, and train their numerical parameters from scratch on each new task. Experiments on a synthetic signal-detection benchmark show that transferring these structures improves generalization over tuned fully connected baselines.

Keywords: inductive bias, generative model

1 Introduction

Choosing an appropriate inductive bias is crucial for generalization [1]. By constraining the hypothesis space to reflect task structure, it can enable effective learning from limited data. Examples include local connectivity and spatial weight sharing in convolutional networks [2], and recurrent state updates with parameters shared across time [3]. Suitable parameter-sharing patterns also enforce equivariance to task-relevant transformations [4].

Selecting suitable structural biases by hand requires domain expertise, particularly when the relevant properties of a new task are not known in advance [5]. This motivates learning such biases from data. Yeh et al. [5] optimize parameter-sharing schemes to discover equivariances, while Zhou et al. [6] meta-learn sharing patterns across related tasks. Learning groups of similar weights has also been studied through soft weight tying [7] and weight-sharing regularization [8]. Atanov et al. [9] learn generative priors from convolutional filters of previously trained networks and use them to guide learning on new datasets. Schürholt et al. [10] learn generative representations of trained networks to sample new numerical weights. Sun et al. [11] extract task-specific subnetworks by iterative pruning and share parameters where these subnetworks overlap. Ando and Zhang [12] learn a shared predictive structure across tasks while allowing task-specific predictors.

More generally, a single task may admit multiple near-optimal structures, and its data alone need not identify which constraints will transfer to other tasks. Baxter [1] provides a theoretical framework for learning a shared hypothesis space across tasks, with generalization bounds for new tasks from the same environment under suitable capacity and sampling assumptions. Our focus is on transferring connectivity and weight-sharing constraints extracted from trained solutions, with numerical values learned from scratch for each new task.

In this paper, we present a generative multi-task learning framework that learns a distribution over network structures from solutions to related tasks. We represent a structure by an indexator that specifies which connections are present and which weights share a numerical value. The corresponding codebook contains the trainable values. We retain candidate structures according to their validation performance after training a fresh codebook on source tasks.

Our central hypothesis is that structural regularities shared across source solutions provide an inductive bias that transfers to new tasks. We model these regularities with an autoregressive distribution over indexators, using MADE [13]. On a new task, the frozen generator proposes structures, whose numerical parameters are trained from scratch and whose performance is assessed on query data.

On a synthetic benchmark adapted from Dynamic Signal Distribution [14], the extracted structures improve generalization over tuned fully connected models and the same connectivity mask without weight sharing. Simple frequency-based proposals achieve similar accuracy to the autoregressive model. A real-image control using input-selection masks shows no advantage over tuned dense baselines.

Our main contributions are as follows:

– We establish sufficient conditions under which structural search finds low-error solutions that dense gradient-based optimization is unlikely to reach within a fixed number of steps.
– A multi-task framework for transferring connectivity and weight-sharing constraints while learning new numerical parameters.
– A procedure for extracting, validating, and autoregressively generating indexators from source solutions.
– An empirical evaluation of structural recovery and transfer, including simple structural baselines and a real-image control.

2 Related Works

TODO

3 Preliminary

A trained network contains information about both the numerical values of its parameters and the relations between them. For example, a predictor may use only a subset of the available connections and share a coefficient across several of them. These relations may remain useful on another task even when the required coefficients change. Learning such constraints from related tasks could therefore help a new predictor generalize from limited data. We evaluate their usefulness through the quality of the solutions obtained after learning new numerical parameters.

3.1 Multi-task learning

Multi-task learning uses information shared by related prediction tasks to improve learning. In our setting, each task τ is a supervised learning problem with its own distribution P_τ over input–target pairs (x, y). Tasks share input and target spaces $\mathcal{X}$ and $\mathcal{Y}$, but may require different predictors. We model their relationship through a task family $\mathcal{T}$ and a distribution $p(\tau)$ describing which tasks occur [1].

We use source tasks $\mathcal{T}_{\text{tr}}$ to learn shared structural information and a disjoint collection of target tasks $\mathcal{T}_{\text{te}}$ to assess its transfer. Both collections are drawn from the same task environment. For each task, a support sample S_τ^{s} provides examples for fitting the numerical parameters. An independent query sample S_τ^{q} provides examples for comparing the resulting predictors [6, 15]. Both samples contain observations drawn independently from P_τ. For a predictor f and a sample S, we measure performance by the average prediction loss

$$\mathcal{L}_S(f) := \frac{1}{|S|} \sum_{(x,y)\in S} \ell(f(x), y). \tag{1}$$

Here, ℓ is the loss for a single example, such as cross-entropy for classification or squared error for regression.

3.2 Network structure

We fix a common network architecture and collect the weights to be structured into a vector $w \in \mathbb{R}^P$, where P is the number of selected weight positions. An integer array $I \in \{0, \dots, K\}^P$, called an *indexator*, assigns each position to an entry of a *codebook* $C_\tau = (C_{\tau,0}, \dots, C_{\tau,K})$. The codebook contains K trainable values and a fixed zero entry. The effective weights are

$$w_p = C_{\tau,I_p}, \qquad p = 1, \dots, P, \qquad C_{\tau,0} = 0. \tag{2}$$

Connections with the same positive index must share a numerical value; index zero disables a connection. Thus, I specifies both connectivity and weight sharing, while C_τ supplies the values for a particular task. A connectivity mask is a special case in which each active connection has its own trainable codebook entry.

For example,

$$I = (1, 2, 1, 0), \qquad C_\tau = (0, 0.7, -1.2) \quad \Longrightarrow \quad w = (0.7, -1.2, 0.7, 0).$$

The first and third weights are tied, and the fourth is fixed at zero. On another task, the same indexator can use different codebook values while preserving these relations. We write f_{I,C_τ,θ_τ} for the resulting predictor, where θ_τ contains any remaining trainable parameters, such as biases. Both C_τ and θ_τ are learned afresh for each task, with I held fixed.

3.3 Good solutions and structural transfer

To judge a structure, we fit its numerical parameters on the support sample using a specified optimizer, initialization rule, and budget of T updates. Let $\widehat{f}_{\tau,I}$ denote the resulting predictor. For an error tolerance ε, we call it a *good solution* if its expected loss on new examples from the task is at most ε:

$$\mathcal{R}_\tau(\widehat{f}_{\tau,I}) := \mathbb{E}_{(x,y)\sim P_\tau}\left[\ell(\widehat{f}_{\tau,I}(x), y)\right] \le \varepsilon. \tag{3}$$

The data available for fitting and the training budget are part of this criterion: a structure is useful when its parameters can be learned well under those conditions.

In practice, the expected loss is usually unknown, so we compare fitted predictors using their query loss. When query data are used to select a candidate, final performance is measured on additional test examples unused in selection. We assess structural transfer across new tasks and fresh initializations to determine how reliably the proposed constraints lead to good solutions.

Several structures may support good solutions, and their usefulness may vary across tasks. Our shared model is therefore a distribution $Q_\phi(I)$ over indexators, with parameters ϕ learned from source tasks. On a target task, this distribution is kept fixed and supplies candidate structures. Each candidate receives newly trained numerical parameters, and the query sample is used to select the resulting predictor.

4 Method

TODO test

5 Experiments

TODO

6 Conclusion

TODO

References

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